Shiva Acharya

Ph.D. Candidate, Electrical and Computer Engineering (Wireless @ Virginia Tech)
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EDUCATION

Ph.D. Candidate, Computer Engineering
Virginia Tech, Blacksburg, VA Spring 2021 - Present
Expected Graduation: Spring 2026

M.S., Computer Engineering
Virginia Tech, Blacksburg, VA Spring 2023

B.S., Electrical Engineering (Summa Cum Laude)
McNeese State University, Lake Charles, LA Spring 2020

HONORS AND AWARDS

Awarded Top Technical Presentation at CCI Graduate Student Summit	2024
Awarded IEEE MILCOM 2024 Student Travel Grant	2024
Awarded CCI SWVA Cyber Innovation Scholar Fund	2024
Chosen as CCI SWVA Cyber Innovation Scholar	2024
Featured on VT News: Rung by rung, graduate students climb toward cybersecurity careers	2024
Awarded Most Technically Enlightening Presentation at *Inaugural CCI Graduate Student Summit*	2023
Spotlight: Featured student at CCI SWVA	2023
Awarded CCI SWVA Cyber Innovation Scholar Fund	2023
Chosen as CCI SWVA Cyber Innovation Scholar	2023
Summa Cum Laude, McNeese State University	2020
Won Design Award at VEXU Robotics Competition, Houston, TX, USA	2020
President's Honor List, McNeese State University	2016–2020

PUBLICATIONS

- **S. Acharya**, S. Li, Y. Wu, N. Jiang, W. Lou, and Y.T. Hou, "A Spectrum-Efficient Solution with Data Rate Guarantees in 5G/Next-G Networks," **Under Review** *IEEE Internet Of Things Journal.*
- **S. Li**, N. Jiang, C. Li, **S. Acharya**, Y. Wu, W. Xie, W. Lou, and Y.T. Hou, "Real-time MU-MIMO Beamforming with Limited Channel Samples in 5G Networks," **Under Review** *IEEE Transactions on Mobile Computing.*
- **S. Acharya**, S. Li, Y. Wu, N. Jiang, W. Xie, W. Lou, and Y.T. Hou, "A New Approach to Tackle Channel Uncertainty with Limited Data Samples," **Under Review** *IEEE Wireless Communications Magazine.*
- **S. Acharya**, S. Li, Y. Wu, N. Jiang, W. Lou, and Y.T. Hou, "Rudra: An Algorithm for Optimizing Spectrum Efficiency with Data Rate Guarantee in Next-G Communications," *Proc. IEEE MILCOM*, Washington D.C., USA, 28 Oct.–Nov. 01, 2024.
- **E. Ghoreishi**, B. Abolhassani, Y. Huang, **S. Acharya**, W. Lou, and Y.T. Hou, "Cyrus: A DRL-based Puncturing Solution to URLLC/eMBB Multiplexing in O-RAN," *Proc. IEEE ICCCN*, pp. 1–9, Kailua-Kona, HI, USA, 29–31 July, 2024.
- **S. Acharya**, S. Li, N. Jiang, Y. Wu, Y.T. Hou, W. Lou, and W. Xie, "Mitra: An O-RAN based Real-Time Solution for Coexistence between General and Priority Users in CBRS," *Proc. IEEE MASS*, pp.

295–303, Toronto, Canada, 25–27 Sept. 2023.

TECHNICAL STRENGTHS

Programming Languages: C/C++, Python, CUDA, Matlab, Java
Software: Visual Studio, Spyder, Matlab/Simulink (5G Toolbox), Eclipse
Skills: Extensive knowledge of wireless networking, algorithm design, and real-time optimization, with a strong foundation in deep learning models such as CNNs, RNNs, GANs, encoder-decoder architectures, DQNs, SAC, and more.

Releveant Courses:

• Digital Communication Advanced Theory & Analysis (ECE 5654) • Information Theory (ECE 5634) • Software Radios (ECE 5674) • Multi-Channel Communication (ECE 6634) • Network Architecture and Protocols (ECE 5566) • Multimedia Networking (ECE 6564) • Spectrum Sharing 5G & IoT (ECE 6504) • Optimization (ISE 5406) • Advanced Machine Learning (ECE 5424) • Introduction to Deep Learning (CS 5814)

PROJECT AND RESEARCH EXPERIENCES

[P6] DRL for Edge User Association (EUA) in Multi-Access Edge Computing (MEC) Summer 2024–Fall 2024
• *DRL Implementation:* Developed and implemented the Deep Deterministic Policy Gradient (DDPG) algorithm to maximize user satisfaction by optimizing the data rate for mobile devices in MEC systems.
• *Parameter Optimization:* Tuned DDPG parameters for convergence, ensuring adaptability to dynamic user arrivals, departures, mobility patterns, and channel conditions, while maintaining scalability to real-world MEC deployments.

[P5] MU-MIMO Scheduler Under CSI Uncertainty Fall 2023–Spring 2024
• *Optimization:* Designed an algorithm to minimize spectrum usage in MU-MIMO systems through resource allocation, rate adaptation, and beamforming.
• *CSI Uncertainty Modeling:* Modeled CSI uncertainty using limited CSI data without assuming known distributions, transforming the original stochastic optimization problem into a deterministic problem.
• *Data Rate Guarantees:* Developed a solution providing probabilistic data rate guarantees for UEs.

[P4] DRL for URLLC/eMBB Multiplexing Fall 2023–Spring 2024
• *DRL Implementation:* Applied the Soft Actor-Critic (SAC) algorithm to dynamically puncture eMBB resources, meeting URLLC latency requirements while minimizing packet loss for active eMBB users.
• *O-RAN Optimization:* Enhanced the SAC algorithm by leveraging multi-timescale control loops in the O-RAN framework to achieve 3GPP URLLC compliance; implemented a feasibility enforcer mechanism in the near-RT RIC to ensure policy convergence.
• *Link-Level Simulation:* Performed 5G link-level simulations using the MATLAB 5G Toolbox to validate algorithm performance.

[P3] Real-Time Algorithm Design for Spectrum Coexistence Summer 2023–Fall 2023
• *Algorithm Design:* Developed a parallel resource allocation algorithm to meet 5G's 1 ms scheduling requirement under numerology 0.
• *Parallel Processing:* Decomposed the problem into a massive number of subproblems, selecting the promising subproblems and solving them in parallel on an NVIDIA Tesla V100 GPU using CUDA C++ platform.
• *GPU Optimization:* Streamlined memory management, optimized thread block allocation, and minimized communication overhead to meet real-time constraints.

[P2] Spectrum Sharing Under CSI Uncertainty Spring 2022–Summer 2023
• *Uncertainty Modeling:* Addressed CSI uncertainty in CBRS spectrum sharing without assuming known channel distributions.
• *Interference Protection:* Developed a small-data approach using limited CSI samples to ensure interference protection guarantees.
• *Resource Optimization:* Optimized resource and power allocation in secondary networks to provide interference protection to the primary network while maximizing system throughput.

[P1] Project Leader, LaACES Ballooning Program Fall 2019–Spring 2020
• *Team Leadership:* Led the NASA-sponsored LaACES project to design and test a telemetry system tracking a payload at altitudes up to 30 km.
• *Payload and Ground Station Design:* Developed a system to capture and transmit images, GPS data, and atmospheric information in real-time to a ground station.

PRESENTATIONS

• Presented research at the *CCI SWVA Graduate Student Summit,* Blacksburg, VA, USA, Dec. 03, 2024.
• Presented poster at the *Young Scholar Workshop Poster Session - IEEE MILCOM*, Washington D.C., USA, Oct. 30, 2024
• Presented poster at the *CCI Symposium*, Richmond, VA, USA, Apr. 15, 2024
• Presented poster at the *Annual CCI Student Researcher Showcase*, Blacksburg, VA, USA, Mar. 22, 2024
• Presented research at the *CCI SWVA Graduate Student Summit,* Blacksburg, VA, USA, Nov. 03, 2023.

TEACHING EXPERIENCE

Teaching Assistant, ENGR 430: Systems and Control	May 2020–Dec. 2020
Teaching Assistant, ELEN 210: Circuits I	Aug. 2019–May 2020
Lab Assistant, ELEN 341: Linear Electronics	Aug. 2019–May 2020
Lab Assistant, ELEN 362: Microprocessing System Design	Aug. 2019–May 2020

PROFESSIONAL SERVICES

Treasurer, IEEE McNeese Student Chapter	Aug. 2019–May 2020
Treasurer, Nepalese Student Association at McNeese State University	Aug. 2019–May 2020
Referee and Judge (Volunteer), VEXU Robotics Competition, LA	Mar. 2019–Feb. 2020

REFERENCES

Prof. Tom Hou: Bradley Distinguished Professor of ECE, Virginia Tech, thou@vt.edu
Prof. Wenjing Lou: W. C. English Endowed Professor of CS, Virginia Tech, wjlou@vt.edu
Prof. Jeff Reed: Willis G. Worcester Professor of ECE, Virginia Tech, reedjh@vt.edu